

Chapter 24

Post Quantum Cryptography

Securing today's data against tomorrow's quantum threats

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24.1 Introduction

The author of this article started off his journey in quantum computing as a student when Caltech open-sourced John Preskill's notes on quantum computing circa 1999. Fast forward to 2019, and he was involved in early field testing of Post Quantum Crypto solution by a major vendor with the largest telecom group in the UK as a partner. Following that, his first paper on a cloud based roll out of hybrid classical-quantum crypto was published in January 2020 [14], to be followed by a book on quantum machine learning (QML), several patents on quantum tech, and research papers.

The author worked for many years in security, networking, automation and machine learning across Europe, with an academic background in Physics and Mathematics. This triggered a natural interest in security aspects and vulnerability of existing encryption systems surrounding the rise of quantum computers.

24.2 Background

Encryption is used to protect the confidentiality of data. It takes the data as an input and in return outputs code or ciphertext that looks as though it has no meaning.

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Only someone with a special key can decipher that code to obtain the original data, and only the intended recipient should have access to that special key. There are two forms of encryption algorithms: Symmetric Key and Asymmetric Key. To understand the difference, imagine the encryption of data to be equivalent to placing the data into a box and locking it, before sending it to the recipient. Therefore, decrypting data would be equivalent to unlocking the box and accessing the data inside.

Symmetric Key encryption is when the same key is used to lock and unlock that box. This is cost-effective (in terms of computation) but carries a risk. The key (called a Pre-shared Key) needs to be sent to the recipient so they can unlock the box, but we must ensure that it is not intercepted during transit. Any hacker who manages to obtain this key would be able to decipher the message and access confidential data. So, the key must be transmitted securely, just as securely as the locked box itself.

This is where Asymmetric Key Encryption is used. The symmetric key itself is encrypted, but this time, the key used to lock this box (called a Public Key) is different to the key used to unlock the box (called a Private Key). The public and private keys are associated to one another through a complex mathematical computation and only the public key is shared.

Today's most commonly used asymmetric key encryption algorithms are based on how difficult it is to break down large numbers into their prime factors. For example, we can easily verify that the multiplication of 3, 5, 11 and 17 is equal to 2805, but it is much harder to start with the number 2805 and break it down into its prime components. As the number gets larger, the problem becomes exponentially harder to solve, even for a computer. In fact, if the number is made up of at least 2048 bits, scientists believe it will take a standard computer so long to solve that it is practically impossible and so this problem is considered to be 'intractable'. Asymmetric key encryption is computationally expensive. It requires a lot of time, memory and processing power and so is not used to encrypt voluminous data itself, rather just the symmetric key that needs to be sent to the recipient.

However, the study of mathematics has evolved and new algorithms that can quickly reverse the factorising problem are being researched and developed. These algorithms would still require a greater amount of processing power than what is available today, but technology is quickly catching up and it is thought that a classical computer with enough power to run these new algorithms will exist in the near future.

Furthermore, a quantum computer would be able to solve this in just a matter of hours, thanks to the parallel processing capability discussed earlier. This would mean the symmetric key could be compromised, so all data that was encrypted using that key would also be compromised. Fortunately, a quantum computer with this level of processing power has not yet been built, and it is not expected to be around for at least another decade.

So, why are we worried? For the everyday user who makes online transactions (which use 2048-bit encryption), it is not a large reason for concern. If a transaction made today is decoded in twenty years' time, there is little chance of much damage being caused as a result. However, if we are sending and receiving confidential data that will still need to remain secure twenty years into the future, then it would definitely

be worth considering the use of post-quantum cryptography. And the trend of HNDL (harvest now, decrypt later) of encrypted data has been viably established via evidence across various geographical locations for the last six to seven years: there are entities who are actually storing encrypted data today, with hopes of decrypting them in future when computers become powerful enough to crack them. Keeping this in mind, organizations such as NIST (National Institute of Standards and Technology), NCSC (National Cyber Security Centre) et al., scientists and researchers have already placed resources into developing post-quantum encryption algorithms and exploring Quantum Key Distribution to deliver the symmetric key.

24.3 So, What Is This PQC Anyway?

Post-Quantum Cryptography (PQC) refers to a new generation of cryptographic algorithms designed to remain secure even against attacks from quantum computers. While today's widely used public-key systems, such as RSA and elliptic curve cryptography, are considered secure against classical computers, they are expected to become vulnerable once large-scale quantum computing becomes practical. PQC addresses this risk by introducing alternative algorithms for encryption, key exchange, and digital signatures that can withstand both classical and quantum attacks.

For organisations, PQC is not only a technical upgrade but also a strategic risk management priority. Sensitive data with long retention periods, such as financial records, personal information, government data, and intellectual property, may already be exposed to “harvest now, decrypt later” (HNDL) threats, where encrypted information is collected today and decrypted in the future when quantum capabilities mature. As a result, many organisations are beginning to assess their cryptographic landscape, build crypto inventories, and introduce crypto-agile architectures that allow algorithms to be replaced without major system redesign.

24.4 Is PQC the Holy Grail of Security Against All Quantum Attacks?

In reality PQC algorithms are purely classical, based on complex mathematical models that are supposed to be “quantum proof” due to their mathematical complexity. However, in reality, the attacks from a quantum computer, if and when that comes, will come from the domain of physics. Since, no known quantum attacks have been detected yet, it is hard to predict whether or not these classical PQC algorithms will actually be fruitful against such attacks.

As an example, several specific NIST post-quantum cryptography (PQC) candidates, including finalists like SIKE and Rainbow, were broken by researchers using conventional, non-quantum, present day ordinary computers. SIKE (Supersingular Isogeny Key Encapsulation) was broken in 2022 by researchers using a single-core PC in roughly 62 minutes, despite it being a top-round candidate for NIST. Rainbow signature scheme broken in February 2022 by Ward Beullens, who demonstrated a

practical key recovery attack on a laptop. However, the primary algorithms selected for standardization, such as CRYSTALS-Kyber, are perceived to be secure against future quantum threats, with NIST releasing the first finalized standards in August 2024.

The other option for security against quantum attacks is **Quantum Key Distribution (QKD)**. Instead of relying on today's complexity of a mathematical problem which could be overcome in the near future, Quantum Key Distribution (QKD) relies on the unchanging principles of quantum mechanics. It is based on the capability of manipulating a photon particle in order to encode data. A unique key code is generated and then shared only between the sender and the recipient. One thing that is quite unintuitive about a quantum particle is that it behaves differently when someone tries to observe the current state and differently when it is not being observed. This idea of being unable to observe the state of the particle is essential in QKD. If a hacker tries to intercept a key being transmitted via QKD, the state of the photon would change, so the recipient would know immediately that it has been tampered with and could terminate all communication. There are however some limitations to QKD, including the fact that it can only operate via satellite or fibre optics (which is what caught the interest of many service providers, such as BT). Whilst using a satellite can allow QKD over huge distances, transmission over commercial fibre keeping the coherence of information at an acceptable level, is currently limited to around 50km-60km; however, this distance can be extended through the use of specific optical repeaters.

Note: Whilst QKD sounds very promising, it is still in the stage of early trials and research, are not quite standardized for usage at a large scale (even though smaller scale, campus deployments are up and running in real world), and therefore the recommended direction, given by the National Cybersecurity Center UK, is as follows:

- Do not endorse QKD for any government or military applications
- Advise against replacing any existing public key solutions with QKD for commercial applications¹

Despite this, there has been strong progress into the creation and testing of QKD networks around the globe. In Vienna, a QKD network has been designed and implemented and what is more is that China and Vienna have shared a key that was distributed via QKD-satellite. The UK has also made huge investments in implementing and testing QKD networks. The author of this article published a defensive article in January 2020 named "System and Architecture of a Quantum Key Distribution (QKD) Service over SDN" [19] along with a few patents on quantum security with quantum tech as well as with hybrid-quantum classical approaches [see references].

¹Source: <https://www.ncsc.gov.uk/whitepaper/quantum-key-distribution>

24.5 PQC: Some Technical Overview

Post-quantum cryptography (PQC) is the transition from quantum-vulnerable public-key cryptography (notably RSA/ECC-based key establishment and digital signatures) to algorithms designed to resist future quantum attacks. As mentioned before, NIST approved the first three PQC standards in August 2024: FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA), providing a standards-based foundation for enterprise planning and implementation.

NIST has announced 3 different standards:

The new accepted PQC standards are designed for two essential tasks for which encryption is typically used: general encryption, used to protect information exchanged across a public network; and digital signatures, used for identity authentication. NIST announced its selection of four algorithms — CRYSTALS-Kyber, CRYSTALS-Dilithium, SpHincS+ and FALCON — slated for standardization in 2022.

- **For general encryption**, used when users access secure websites, NIST has selected the CRYSTALS-Kyber algorithm. Among its advantages are comparatively small encryption keys that two parties can exchange easily, as well as its speed of operation.
- **For digital signatures**, often used when we need to verify identities during a digital transaction or to sign a document remotely, NIST has selected the three algorithms CRYSTALS-Dilithium, FALCON and SPHINCS+ (read as “SpHincS plus”). Reviewers noted the high efficiency of the first two, and NIST recommends CRYSTALS-Dilithium as the primary algorithm, with FALCON for applications that need smaller signatures than Dilithium can provide. The third, SPHINCS+, is somewhat larger and slower than the other two, but it is valuable as a backup for one chief reason: It is based on a different math approach than all three of NIST’s other selections.

Three of the selected algorithms are based on a family of math problems called structured lattices, while SPHINCS+ uses hash functions. The additional four algorithms still under consideration are designed for general encryption and do not use structured lattices or hash functions in their approaches.

While there have been no substantive changes made to the standards since the draft versions, NIST has changed the algorithms’ names recently to specify the versions that appear in the three finalized standards, which are:

- Federal Information Processing Standard (FIPS) 203, intended as the primary standard for general encryption. Among its advantages are comparatively small encryption keys that two parties can exchange easily, as well as its speed of operation. The standard is based on the CRYSTALS-Kyber algorithm, which has been renamed ML-KEM, short for Module-Lattice-Based Key-Encapsulation Mechanism.
- FIPS 204, intended as the primary standard for protecting digital signatures. The standard uses the CRYSTALS-Dilithium algorithm, which has been renamed ML-DSA, short for Module-Lattice-Based Digital Signature Algorithm.

- FIPS 205, also designed for digital signatures. The standard employs the Sphinx+ algorithm, which has been renamed SLH-DSA, short for Stateless Hash-Based Digital Signature Algorithm. The standard is based on a different math approach than ML-DSA, and it is intended as a backup method in case ML-DSA proves vulnerable.

Similarly, when the draft FIPS 206 standard built around FALCON is released, the algorithm will be dubbed FN-DSA, short for FFT (fast-Fourier transform) over NTRU-Lattice-Based Digital Signature Algorithm.

While the standard is in development, *NIST encourages security experts to explore the new algorithms and consider how their applications will use them, but not to bake them into their systems yet, as the algorithms could change slightly before the standard is finalized.*

One of the most important considerations for PQC deployment is migration. NIST NCCoE guidance emphasizes that migration to PQC from our existing encryption standards requires organizations to identify where quantum-vulnerable public-key cryptography is used across hardware, software, and services, and to build prioritized migration roadmaps. This is not a one-time library upgrade; it is a multi-year program involving discovery, architecture updates, interoperability testing, phased deployment, supplier coordination, and operational monitoring.

This engineering-focused report provides a practical framework with migration phases, test matrices, rollout guardrails, runbooks, KPIs, and dual-lens guidance for executives and engineering teams.

Callout. The decision is not whether PQC matters - it is whether the organization will execute a controlled, evidence-based migration early enough to avoid operational disruption later. NIST standards are available; planning and pilots should begin now.

Technical Acumen. Treat PQC as a platform modernization program: crypto inventory (CBOM), protocol support, interoperability testing, telemetry, rollout automation, and rollback readiness are as important as algorithm selection.

24.6 Why PQC Matters Now

24.6.1 Migration lead time is the immediate challenge

Even if the exact date of cryptographically relevant quantum computers remains uncertain, cryptographic transitions are slow. Organizations need time to identify cryptography embedded across third-party products, devices, protocol stacks, PKI, code-signing workflows, and customer/partner integrations. To this end, NIST has rolled out a migration guideline.

The dominant short-term risk for many enterprises is migration unreadiness rather than immediate cryptanalytic break. Delayed planning increases the chance of rushed deployments, compatibility failures, and uncontrolled exception sprawl.

Callout. Migration unreadiness is a business resilience risk. Budgeting for discovery, testing, and supplier engagement early reduces future operational disruption.

24.6.2 Harvest Now, Decrypt Later (HNDL) exposure

Adversaries may collect encrypted traffic or archives today and attempt decryption later if the underlying public-key mechanisms become vulnerable. This is especially relevant for long-lived sensitive data, regulated records, intellectual property, and software trust chains.

24.6.3 What is most exposed first

PQC migration primarily targets public-key mechanisms used for key establishment and digital signatures. Typical high-priority domains include TLS/VPN handshakes, PKI and certificates, code signing, firmware signing, secure boot, and identity/authentication systems.

Although some symmetric cryptography such as RSA below 2048 and ECC are deemed vulnerable, the overall domain remains comparatively resilient in the near term when appropriately parameterized, but policy and architecture reviews and updates are still needed.

24.7 NIST PQC Standards and How They Work

24.7.1 Current NIST baseline

NIST approved FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) in 2024. FALCON remains selected for future standardization (FIPS 206 in development).

24.7.2 Naming translation for teams

- CRYSTALS-Kyber → ML-KEM
- CRYSTALS-Dilithium → ML-DS
- SPHINCS+ → SLH-DSA (standardized form)
- FALCON → future FIPS 206 (in development)

Organisations use the FIPS names consistently in architecture reviews, procurement, and audit artifacts, reducing confusion when teams reference older competition-era names. NSA's CNSA FAQ explicitly clarifies that ML-KEM and ML-DSA are the fully specified NIST standards corresponding to those earlier submission names.

24.7.3 ML-KEM (FIPS 203): conceptual behavior and deployment relevance

ML-KEM is a key encapsulation mechanism (KEM). At a high level, one party generates a KEM public/private key pair; the peer encapsulates a shared secret to the public key, producing a ciphertext and shared secret; the private key holder decapsulates and derives the same shared secret. The resulting shared secret is used to derive symmetric session keys.

- Deployment relevance: TLS/QUIC-like session establishment
- VPN key establishment
- Service-to-service secure channels
- Secure messaging/session setup

Technical Acumen. KEM integration changes more than algorithm names. It can affect handshake payload sizes, CPU cost, retry/timeout behavior, and observability requirements.

24.7.4 ML-DSA and SLH-DSA

ML-DSA (FIPS 204) and SLH-DSA (FIPS 205) are PQC digital signature standards used for integrity, authenticity, and non-repudiation use cases, subject to ecosystem support. Typical uses include code signing, signed metadata, firmware signing, and identity artifacts.

24.7.5 Standards and deployability

Production rollout depends on standards, protocol support, libraries, HSM/TPM support, certificate, client/server compatibility, and operations readiness. Fig. 1 below shows a protocol level conceptual schematic for KEM-based processes.

24.8 Engineering Migration Roadmap (3–5+ Year Program)

A practical transition to post-quantum security is typically phased. It begins with discovery and risk assessment, followed by pilot implementations using hybrid cryptography, where classical and post-quantum methods are used together, and then expands into broader enterprise adoption. In regulated sectors such as finance and government, PQC is increasingly viewed as a foundational component of long-term cyber resilience, Zero Trust architecture, and compliance readiness.

Any upgrade to PQC from existing crypto will require thorough preparation and diligent planning. The exact timeline depends on sector, legacy footprint, and vendor readiness. Public milestone models such as the UK NCSC's 2028/2031/2035 guidance

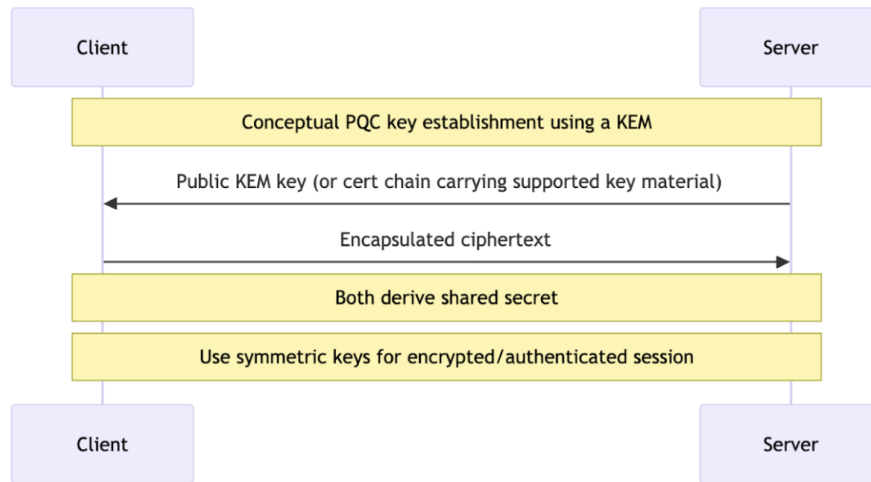


Figure 24.1: High-level conceptual flow for KEM-based key establishment. This is an explanatory schematic, not a protocol specification.

can be used as external planning anchors while preserving an enterprise-specific phased roadmap.

24.8.1 Crypto-agility

Prior to upgrading to PQC, it is strongly advised that the systems go through a thorough cryptographic inventory and define protocols for crypto-agility. **An crypto agile system** is one that automatically adjusts to any security threats or compliance lapse in the digital environment, as for example, may happen during mergers of two companies with two different security standards.

Crypto-agility is formally defined as the ability of a system to adapt its cryptographic mechanisms without requiring major redesign, disruption, or prolonged downtime. As cryptographic standards evolve and new threats emerge, organisations need architectures that can replace algorithms, keys, certificates, and protocols quickly and safely. The core design principle of crypto-agility is to avoid hard-coding specific algorithms into business applications. Instead, cryptographic functions should be abstracted through reusable services, APIs, or policy-driven platforms so that encryption, signing, and key exchange methods can be updated with minimal impact on the wider system. This kind of policy enforcement is of optimal importance in environments that contain sensitive data, such as Financial organisations or Federal government.

Effective crypto-agility also depends on centralised governance and clear policy control. Cryptographic choices should be defined by enterprise policy rather than individual developers or isolated teams. This includes approved algorithms, key sizes, certificate profiles, rotation schedules, and deprecation rules. Strong key and trust management is equally important, with lifecycle controls for key generation, storage, rotation, revocation, and destruction. Systems should also be designed for interoperability and phased migration, allowing classical and next-generation cryptography to coexist

during transition periods. This is especially important in post-quantum migration, where hybrid approaches help maintain compatibility while reducing future risk.

Finally, *crypto-agile systems must be observable, testable, and auditable*. Organisations need visibility into where cryptography is used, which algorithms are active, and which systems are exposed to legacy or have weakened controls. Logging, inventory, and continuous monitoring make it possible to assess risk and respond quickly when standards change. In this sense, crypto-agility is not only a technical design choice but a resilience strategy: it enables security teams to respond to new threats, regulatory requirements, and technology shifts without rebuilding the enterprise each time cryptography needs to evolve. A general phased roadmap for a PQC migration is shown in Fig. 2 below.

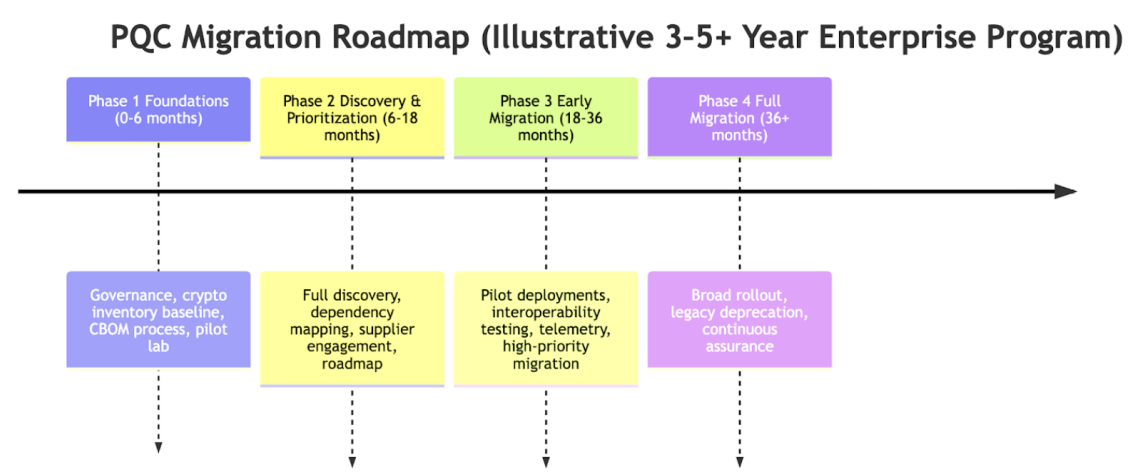


Figure 24.2: Example phased migration roadmap aligned to enterprise delivery cadence, dependent on timelines to asset complexity and supplier readiness.

Fig. 3 below depicts a broader view of phased PQC roll-out as per NIST guidance

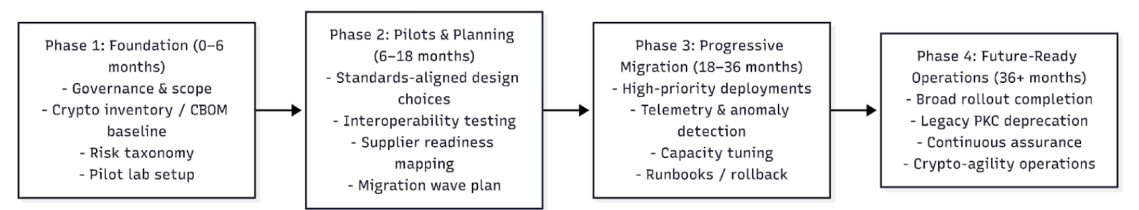


Figure 24.3: Illustrative enterprise roadmap for phased PQC migration. Actual timelines depend on asset complexity, ecosystem readiness, and vendor support.

24.9 Machine Learning (ML) for PQC Deployment, Monitoring, and Capacity Control

ML is most valuable in operational support for PQC migration: discovery and inventory enrichment, telemetry anomaly detection, compatibility prediction, rollout prioritization, and capacity forecasting. This is focused on solving one major problem in today's networks: Crypto usage is scattered across code, configs, appliances, and managed services.

Machine learning can play an important supporting role in Post-Quantum Cryptography (PQC), not by replacing cryptographic standards, but by helping organisations manage the complexity of discovery, migration, and continuous monitoring. As enterprises prepare for the transition to quantum-resistant algorithms, one of the biggest challenges is identifying where current cryptography is used across applications, infrastructure, certificates, protocols, and data flows. Machine learning can assist by analysing code, configurations, network telemetry, and security logs to detect cryptographic usage patterns, classify algorithms, and build a more complete cryptographic inventory. This helps organisations understand which systems rely on vulnerable classical cryptography and where post-quantum migration should be prioritised.

Machine learning is also valuable in risk assessment and operational monitoring. By correlating factors such as data sensitivity, exposure level, retention period, and current algorithm strength, ML models can help score systems according to post-quantum migration urgency. During rollout, machine learning can support anomaly detection by identifying unusual handshake failures, compatibility issues, or unexpected changes in cryptographic behaviour across hybrid environments where classical and post-quantum methods coexist. However, its role should remain carefully bounded: machine learning should inform discovery, prioritisation, and monitoring, while final cryptographic decisions remain governed by approved policy, standards, and human review. In this way, machine learning strengthens PQC programmes by improving visibility, scalability, and resilience without replacing formal security controls. Figure 4 below shows ML-Enabled PQC Rollout Control Loop.

In summary, main ML use cases for PQC involve the following:

- Repository/config classification for crypto indicators
- Protocol/cipher pattern detection in telemetry
- Certificate metadata clustering
- NLP on tickets/docs for hidden dependencies
- Improved CBOM completeness
- Reduced manual discovery time
- Audit evidence generation
- Capacity planning
- Engineering

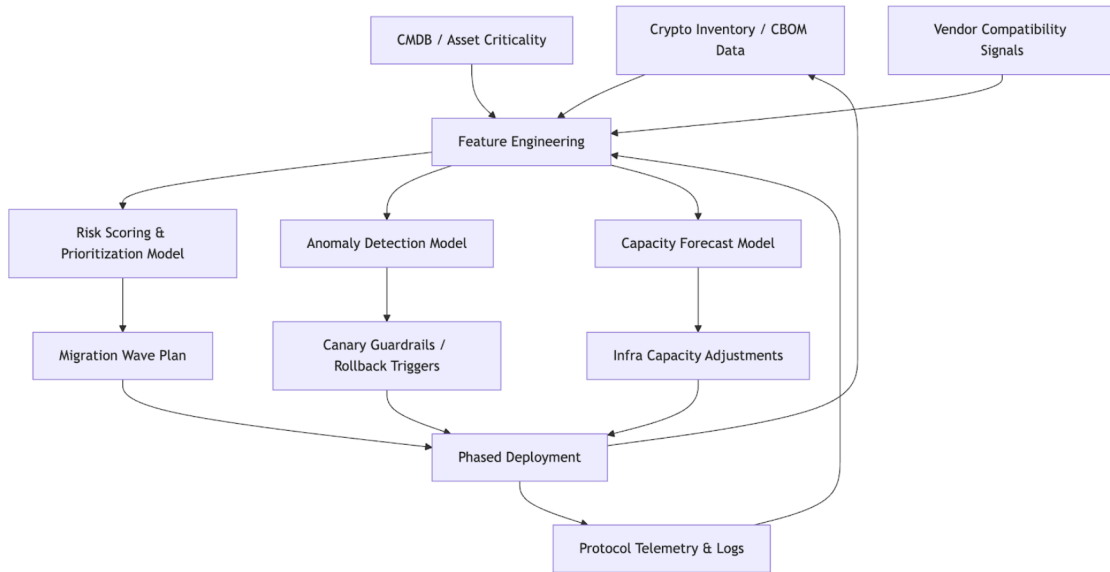


Figure 24.4: Example control loop showing how classical ML can support PQC migration planning and safe phased rollout operations.

24.9.1 Performance and implementation considerations (including GPU acceleration/cuPQC)

Why performance matters

PQC migration is not only a security project. It can affect handshake CPU costs, connection setup latency, message sizes, memory usage, hardware acceleration assumptions just to name a few. This is where NVIDIA’s PQC stack becomes absolutely critical.

NVIDIA’s cuPQC SDK documentation describes it as a high-performance SDK for cryptographic applications on GPUs, and its feature docs position cuPQC-PK as GPU-accelerated implementations of NIST-standardized PQC algorithms with batched operations and device-side integration. Following defines the way ecosystem is adapting to performance concerns:

- high-throughput services,
- cloud-scale termination workloads,
- telecom / infra-heavy deployments,
- research benchmarking.

GPU acceleration (e.g., NVIDIA cuPQC) is one emerging option for organizations that need high-throughput PQC operations or batched cryptographic workloads, but it should be treated as an optimization path rather than a prerequisite for migration.

24.10 QML for PQC: Opportunities and Limits

PQC migration today is driven by standards, software engineering, interoperability, and operations. Even though officially quantum machine learning (QML) is treated as an R&D track, not a prerequisite for enterprise PQC migration, research in the area has picked up considerably in recent times. There has been some works on quantum-enhanced intrusion detection and PQC, albeit, existing studies remain largely performance-focused, with limited analysis of geometric interpretability, stability, and operational robustness.

The author of this article published a book in 2021 called “Quantum Machine Learning: An Applied Approach”, addressing general applied QML as well as security aspects that can be addressed via such methods. Since then research has picked up and QML for cybersecurity and PQC is a popular area for research. An example is a recent focus of the author where a hybrid classical-quantum autoencoder (HCQAE) architecture was investigated to analyse CIC DDoS data from University of New Brunswick.²

Anomaly detection [15, 16, 17, 18] is central to cybersecurity and cyber-physical systems, where rare events must be identified without labelled attack data. Classical autoencoders detect anomalies via reconstruction error and quantum variants have been explored through variational classifiers and optimization-based methods [16, 17, 18].

Quantum anomaly detection remains comparatively underexplored. Existing work includes quantum kernels, variational classifiers, and generative quantum models for detecting distributional shifts. Yet it is largely unresolved whether anomalies appear as structured displacement in quantum state space or whether apparent gains reflect unstable dynamics rather than robust representations. These are exciting areas where QML can be very useful in recent future given NVIDIA’s focus on making quantum AI simulation reachable to industry via HPC boost.

Plausible applications of QML for PQC:

1. Optimization experiments for rollout sequencing
2. Anomaly detection research on specialized telemetry representations
3. Benchmarking hybrid classical/quantum workflows

Callout. QML should be funded as innovation if it aligns with R&D strategy, but do not gate PQC migration readiness on it.

24.11 Risks, Pitfalls, and Recommendations

In every technology roll-out, more so for new technologies, there are always risks and caveats to be aware of.

²<https://www.unb.ca/cic/datasets/ddos-2019.html>

24.11.1 Common mistakes to avoid

1. Treating PQC as a simple library upgrade
2. Waiting for “all standards to be final everywhere”: NCSC and NSA both indicate the transition is multi-year and requires early planning, not last-minute action.
3. Starting pilots without a crypto inventory: No crypto inventory means no CBOM. To prioritize migration, identification of where vulnerable cryptography lives is of paramount importance.
4. Skipping interoperability and telemetry testing
5. Using hybrid as a permanent architecture by default: Hybrid approach can help transition, but NSA notes the complexity and extra long-term transition burden it can introduce.
6. Ignoring supplier and partner dependencies.
7. Lacking rollback criteria and rehearsed runbooks.

24.12 Conclusion

PQC migration is now a practical engineering and governance program, not a speculative future exercise. NIST’s approval of FIPS 203/204/205 provides a standards baseline for planning and pilots, while NIST NCCoE and NCSC guidance make clear that early discovery, interoperability testing, and phased execution are essential.

For executives, success means funding and governing a crypto-agility transformation. For engineering teams, success means delivering inventory, testability, telemetry, staged rollout control, and reliable rollback. ML can materially improve migration operations; QML remains exploratory.

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